

Academics Can Contribute to Domain-Specialized Language Models

Mark Dredze Genta Indra Winata Prabhanjan Kambadur Shijie Wu
Ozan Irsoy Steven Lu Vadim Dabravolski David S Rosenberg
Sebastian Gehrmann

Abstract

Large language models (LLMs) have demonstrated impressive performance across a wide range of tasks, but training state-of-the-art models requires substantial computational and financial resources, placing them largely out of reach for academic institutions. This has led to concerns that academic researchers may be excluded from meaningful participation in the development of cutting-edge LLMs. In this paper, we argue that academics can still play a critical role by focusing on domain-specialized LLMs. We discuss the motivations for domain-specific models, outline strategies for developing and evaluating them within academic constraints, and highlight opportunities for impactful research that does not require industrial-scale resources.

1 Introduction

Large language models (LLMs) have rapidly transformed natural language processing (NLP), achieving remarkable performance across a broad range of tasks. Models such as GPT-3 and its successors have demonstrated strong few-shot and zero-shot capabilities, enabling applications that were previously out of reach. These advances have been driven by scaling model size, training data, and computational resources.

However, the increasing scale of state-of-the-art LLMs has also created a widening gap between industry and academia. Training frontier models requires massive computational infrastructure and financial investment, often available only to large technology companies. As a result, academic researchers may feel excluded from contributing to the development of cutting-edge LLMs.

Despite these challenges, we argue that academia can continue to play a vital and distinctive role in LLM research. Rather than competing directly with industry on general-purpose foundation models, academic researchers can focus on domain-specialized LLMs tailored to specific areas of expertise. These models can provide high value while remaining feasible within academic resource constraints.

In this paper, we outline the case for domain-specific LLMs, discuss how they can be developed and evaluated in academic settings, and highlight research directions where academics can make meaningful contributions.

2 LLMs: A Brief History

LLMs: A Brief History

While modern LMs date back to Jelinek (1976), we summarize very recent history to describe the current environment. In the wake of the popularization of neural word embeddings by word2vec (Mikolov et al., 2013), contextualized representations of language as features for supervised systems

were realized by ELMo (Peters et al., 2018) followed by BERT (Devlin et al., 2019; Liu et al., 2019). BERT and subsequent models became the base models for supervised systems utilizing task-specific fine-tuning and continued pre-training for new domains (Gururangan et al., 2020), e.g., for clinical tasks ELMo (Schumacher and Dredze, 2019) and clinicalBERT (Huang et al., 2019).

Parallel work utilized transformers for autoregressive LLMs, resulting in GPT (Radford et al., 2018), GPT-2 (Radford et al., 2019), BART (Lewis et al., 2020a; Liu et al., 2020), CTRL (Keskar et al., 2019), T5 (Raffel et al., 2020; Xue et al., 2021), and XGLM (Lin et al., 2021). These models had some few-shot capabilities, but they could each be adapted (fine-tuned) for a specific task of interest. Some models were available to academics, though training a new model was beyond reach for many.

GPT-3 (Brown et al., 2020) greatly increased model size and changed our understanding of LLMs. Impressive in-context (few-shot) learning pushed the idea that a single large model could solve a wide range of tasks. While the cost of resources meant training was restricted to a few groups, work focused on training bigger models (Chowdhery et al., 2022; Anil et al., 2023; Zhang et al., 2022; Touvron et al., 2023a; Rae et al., 2021). While only a few could train large models, many studied how best to use them: prompt engineering (Liu et al., 2023), prompt tuning (Han et al., 2022; Wei et al., 2022), evaluation (Liang et al., 2022), among many other topics. Commercial LLM APIs, and eventually open source models (Zhang et al., 2022; Workshop et al., 2022; Touvron et al., 2023a,b; Groeneveld et al., 2024), facilitated this work. Ignat et al. (2024) noted the massive research shift to LLMs reflected in Google Scholar citations. Subsequent work in instruction tuning (Ouyang et al., 2022) and fine-tuning (Wei et al., 2022; Chung et al., 2022; Longpre et al., 2023) have further centralized research around general-purpose models. Many consider fine-tuning for specific applications to be obsolete: why would you tune a model for a specific task when you can tune a single model to do well on all tasks?¹

Despite this view, multiple domain-specific LLMs have demonstrated that domain-specific data leads to models that outperform much larger models (Wu et al., 2023; Taylor et al., 2022). Med-PaLM has shown that adapting even giant LLMs to a specific domain leads to vastly increased performance (Singhal et al., 2022, 2023).² Furthermore, the release of LLaMA (Touvron et al., 2023a) led quickly to Alpaca (Taori et al., 2023) and a wave of new fine-tuned versions of LLaMA for specific tasks. This trend strongly indicates that domain-specific models, especially for constrained sizes, are still highly relevant.

To be clear, our concern is not with closed models, which play an important role in the model ecosystem. Models range from full to limited to no access, with some closed models providing incredibly detailed information (Hoffmann et al., 2022; Rae et al., 2019; Wu et al., 2023) and others providing none (Achiam et al., 2023). Our lament over this focus on general models, either open or closed, is that it draws attention away from work on task- and domain-specific models and evaluations. Academics have become product testers, instead of focusing on tasks where they can play a unique role. Moreover, existing academic benchmarks increasingly serve a reduced purpose for commercial models; we are hill-climbing on benchmarks without a way to ensure existing LLMs have not been trained to excel on these benchmarks (Dodge et al., 2021). Furthermore, we rely on benchmarks in place of deep engagement with an application and its stakeholders.

¹Distillation for task-specific models remains popular if smaller models are desired (Hsieh et al., 2023).

²We acknowledge that the biomedical domain is a rapidly developing area, and GPT-4 without fine-tuning was reported to surpass Med-PaLM 2 (Nori et al., 2023).

3 The Need for Domain-Specific LLMs

The Need for Domain-Specific LLMs

In general, web data does not reflect the needs of all NLP systems. Historically, the community has developed systems for specialized domains such as finance, law, bio-medicine, and science. Accordingly, there have been efforts to build LLMs for these domains (Wu et al., 2023; Taylor et al., 2022; Singhal et al., 2022; Bolton et al., 2023; Luo et al., 2022; Lehman et al., 2023; García-Ferrero et al., 2024). We need a deep investment in how best to develop and evaluate these models in partnership with domain experts. How should we best integrate insights gained from the development of general-purpose models with these efforts? We propose several research directions.

How can general-purpose models inform domain-specific models? Building domain-specific models should benefit from insights and investments into general-purpose models. There are several strategies: training domain-specific models from scratch (Taylor et al., 2022; Bolton et al., 2023), mixing general and domain-specific data (Wu et al., 2023), and fine-tuning existing models (Singhal et al., 2022, 2023). Focusing on domain-specific needs, applications, and knowledge with guidance from topic experts will benefit us in acquiring a better model for specific NLP tasks. Which approach yields the best results for task performance and overall cost?

What is the role of in-context learning and fine-tuning? Both LIMA (Zhou et al., 2023) and Med-PaLM (Singhal et al., 2022) use a small number of examples to tune a model. With expanding context size, we may soon rely entirely on in-context learning (Petroni et al., 2020). The interaction between in-context learning and parameter updates remains an open question, particularly for specialized domains where data may be scarce.

How should we evaluate domain-specific LLMs? Evaluation in specialized domains requires collaboration with subject matter experts and the development of high-quality benchmarks that reflect real-world use. Many existing benchmarks were not designed with domain-specific applications in mind, and new evaluation paradigms may be needed to measure utility, safety, and robustness in context.

4 Evaluation of Domain-Specific Models

Evaluation of Domain-Specific Models

Developing domain-specific LLMs requires rigorous evaluation that reflects the needs of the target domain. While general-purpose benchmarks provide useful comparisons across models, they often fail to capture the nuances and requirements of specialized applications. Domain-specific evaluation must therefore go beyond leaderboard performance and consider task relevance, reliability, and alignment with domain standards.

Collaboration with domain experts is essential in constructing meaningful evaluation datasets and metrics. Experts can help identify realistic use cases, define acceptable error rates, and assess qualitative aspects such as reasoning quality and factual consistency. Without such collaboration, models risk optimizing for superficial benchmark gains rather than practical utility.

Another challenge lies in data contamination and benchmark leakage. As commercial LLMs are trained on increasingly large and opaque datasets, it becomes difficult to ensure that evaluation benchmarks have not been included in training data. This concern is especially pronounced for widely used academic benchmarks. For domain-specific models trained within academic settings, transparent data documentation and careful curation can help mitigate these issues.

Finally, evaluation should account for deployment considerations, including robustness, safety, and fairness within the domain context. A model that performs well on isolated test sets may

still fail under real-world conditions. Ongoing monitoring and iterative evaluation are therefore necessary components of responsible domain-specific model development.

5 The Role of Academics

The Role of Academics

A focus on general-purpose LLMs has forced academics to work with large base models and perhaps, shifted the focus to solve problems of immediate industrial interest. Many academics feel excluded from current research trends (Ignat et al., 2024) and the academic and industry relationship is changing (Littman et al., 2022). Shifting attention back to domain-specific applications emphasizes areas where academics hold an advantage: partnerships with domain experts to invest in specific tasks, and consideration of broader societal needs.

Developing domain-specific models requires domain expertise and universities are diverse academic environments that house experts in many domains. Collaborations with these experts can identify data sources, tasks, and challenges important within each domain. Furthermore, these collaborations are the best avenues for better alignment of evaluations with use cases (Winata et al., 2024), and can support the development of proper metrics. These collaborations are necessary to explore wide open interdisciplinary topics, such as models for protein structure prediction (Tunyasuvunakool et al., 2021; Vig et al., 2021) and games as proxies for reasoning (Silver et al., 2016; Agostinelli et al., 2019; Schrittwieser et al., 2020). This includes developing domain-specific resources, which require domain experts to properly design and construct the datasets. Further, areas where industry underinvests are those where academics could focus attention. For example, low-resource languages are not served by a general-purpose multilingual LLM, nor will we reasonably have enough data to support current LLM training methods. Dialects and variations in languages are still wide open topics (Aji et al., 2022; Winata et al., 2023; Nicholas and Bhatia, 2023).

General-purpose LLMs are unlikely to solve problems in many important domains, with many open research problems that can only be solved by domain-specific approaches. Focusing on domain-specific knowledge will benefit us in acquiring a better model and developing application strategies more aligned with how humans learn domain-specific knowledge (Tricot and Sweller, 2014). For many interdisciplinary areas, subject matter experts are essential, and the problems must be defined clearly. The first pass from an LLM is often impressive, but it hides the trenches and areas where things are most interesting. We need a renewed focus on developing and evaluating domain-specific models and applications, an area where academics can play a leading role. Let us not be distracted by claims that a single model solves all tasks, and instead deeply explore and understand the needs and challenges of specific domains.

6 Limitations

Limitations

The literature that we explored in this opinion paper is limited to the area of LLMs. We study the history of LLMs from the literature on word embeddings, encoder-only, and generative transformers to the latest advancement of API-based LLMs.

7 Ethics Statement

Ethics Statement

Our work does not include any experiments or use of data. No potential ethical issues in this work.

8 References

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